Galaba Vamsi

Robotics Engineering Intern | B.Tech. Mechatronics, IIT Bhilai, May 2027 galabavamsi12@gmail.com
galabavamsi.com | linkedin.com/in/galabavamsi | github.com/Galabavamsi

SUMMARY

Mechatronics student who works across the robot stack: ROS 2 and MuJoCo simulation, kinematics and control, robot training data, and safe rollout of robot policies. 15 merged PRs at 3 YC robotics startups.

EDUCATION

B.Tech. in Mechatronics, Indian Institute of Technology (IIT) Bhilai Aug 2023 – May 2027
Coursework: Optimal Control, Advanced Control Theory, Deep Learning for Computer Vision, Digital Image Processing

EXPERIENCE

Co-founder, Robotropic (physical-AI startup) | robotropic.com Oct 2026 – Present
- Building Harness to roll out new robot policies (VLA checkpoints) cell by cell, with shadow-run scorecards and rollback.
- Lead its factory simulator and change gateway, which checks every human or AI-agent action against safety limits.

Research Intern, IIT Bhilai | Advisor: Dr. Arzad Alam Kherani May 2025 – Jul 2025
- Built RIS-SIM, an open-source Python emulator for smart radio surfaces (RIS); co-author, IEEE INDICON 2025.

ROBOTICS PROJECTS

Whiteboard Cleaning Robot (team project) 2025
ROS 2 Jazzy, Gazebo Harmonic, URDF/Xacro, ros2_control, rclpy
- 3-axis Cartesian gantry driven through ros2_control; raster cleaning path with 10 cm overlap over a 90 x 90 cm board.

SCARA Pick-and-Place Arm 2026
MuJoCo, Python
- Turns text commands (“sort red, blue, green”) into pick-and-place plans executed with analytic inverse kinematics.

VLA Policy Fine-Tuning on Robot Arms (π0.5) 2026
Python, PyTorch | private work
- Fine-tuned and trained π0.5 vision-language-action policies with custom training runs for robot-arm manipulation.

Quarter-Car Active Suspension Control (Advanced Control Theory) 2026
MATLAB/Simulink, C, QEMU
- Reproduced an IEEE TSMC 2022 controller within 6%; found an 88 Hz mode that destabilizes its 50 Hz version.
- Ported the controller to C for a Cortex-M4 (tested in QEMU).

Pneumatic Sorting Digital Twin 2026
Unity, C# | github.com/Galabavamsi/pneumatic-sorting-digital-twin
- 3 electropneumatic kits with PLC-style I/O; fault injection (E-stop, air leak, stuck actuator, failed sensor) with replay.

World Model for Robot Manipulation: JEPA on LeRobot Push-T Aug 2026
PyTorch | write-up: galabavamsi.com/blog
- 1.9M-parameter world model on 160 Push-T episodes: latents learned the block's state, but planning scored 0; published why.

OPEN-SOURCE CONTRIBUTIONS

HFlow, Hebbian Robotics (YC S26): 8 merged PRs, 7 issues filed (all fixed) Sep 2026 – Oct 2026
Robot training-data SDK | Python, Apache Arrow, MCAP, LeRobot | github.com/Hebbian-Robotics/hflow
- Fixed robot-log export silently dropping fields missing from the first message (#657); stopped 12-bit depth loss (#400).

Inspect Robots, Robocurve (YC S26): paper co-author; 2 merged PRs, 3 in review Sep 2026 – Oct 2026
Evals for LLM and VLA robot policies | Python | github.com/robocurve/inspect-robots
- Checkpoint/resume for long robot evals; safety aborts and hardware faults are never retried; 80 tests (#514, in review).

Cerulion (YC S24), open-source robot operating system: 5 merged PRs, 4 issues filed Oct 2026
Rust | github.com/cerulion-inc/cerulion
- Made ros2 attach --dry-run work outside a workspace: new Rust module, 217 lines, 8 tests (PR #216).

SKILLS

Robotics: ROS 2 (rclpy, ros2_control, URDF/Xacro), Gazebo, MuJoCo, Unity, MCAP, LeRobot, sensor integration
Control and perception: Inverse kinematics, LQR, Kalman filter, MATLAB/Simulink, OpenCV, PyTorch, VLA fine-tuning
Languages and tools: Python, C, C++, Rust, TypeScript; Linux, Git, Docker, GitHub Actions, pytest

ACHIEVEMENTS AND HARDWARE BUILDS

- Tech Lead, Inter IIT Tech Meet 14.0: 5th of 23 IITs (2025); Top 8 of 4,500+ teams, RAKSHAM hackathon (2026)
- Built a Mars rover for a NASA competition, an EEG-controlled wheelchair prototype and a sensor-based smart blind stick
- Top 1% in JEE Advanced; Prime Minister's Scholarship Scheme scholar (2024); 5th of 232 teams, WEAR Challenge (Kaggle)

PUBLICATIONS

- Co-author, Inspect Robots: Evaluating the Capabilities and Safety of Embodied AI, arXiv:2610.06306; CoRL 2026 workshop
- Co-author, 2 RIS papers: IEEE INDICON 2025 and *Ad Hoc Networks* (Elsevier, 2026)